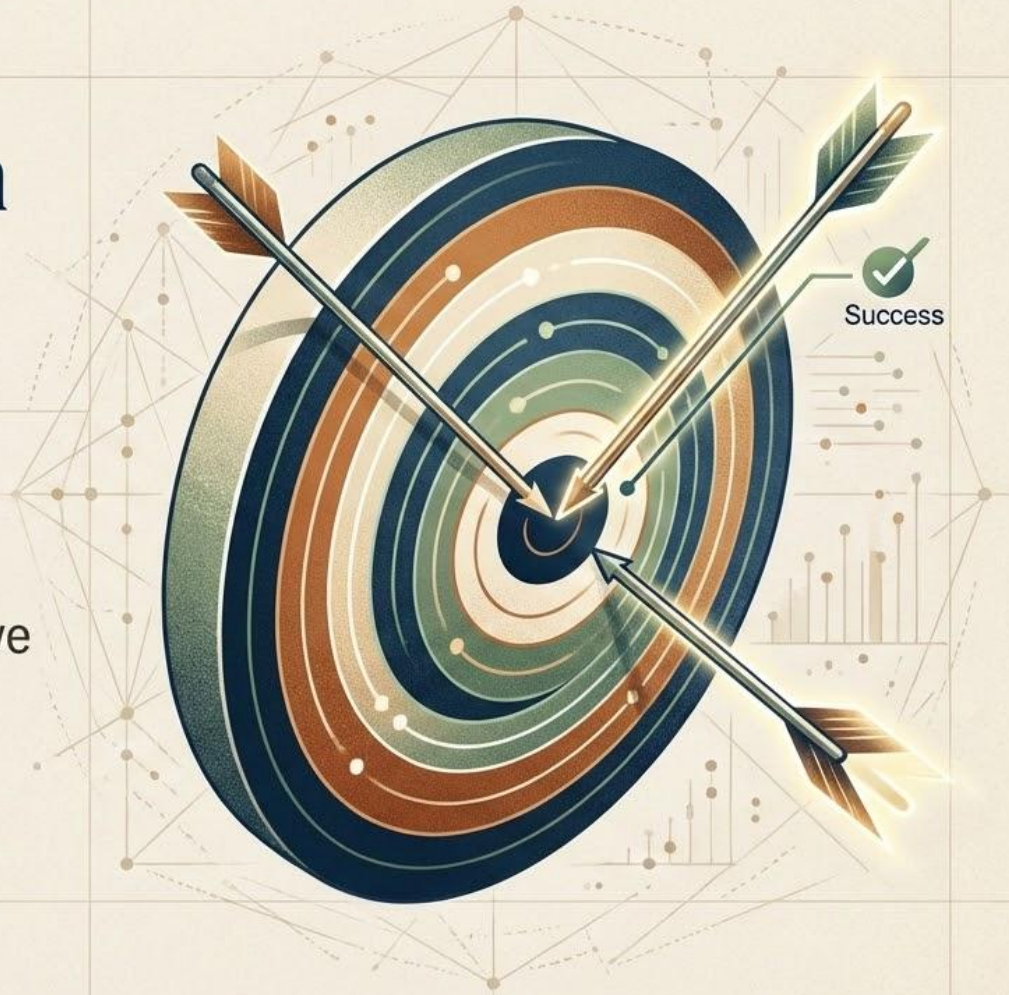


Evaluation Metrics in Machine Learning

A strategic guide to measuring **accuracy**, ensuring **reliability**, and building real-world **trust**. How do we know our model is actually good?

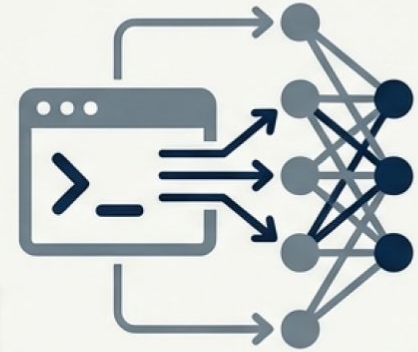




The Diagnosis.

- Correct Diagnosis = Trust
- Wrong Diagnosis = Risk

The Trust Gap: Why We Measure.

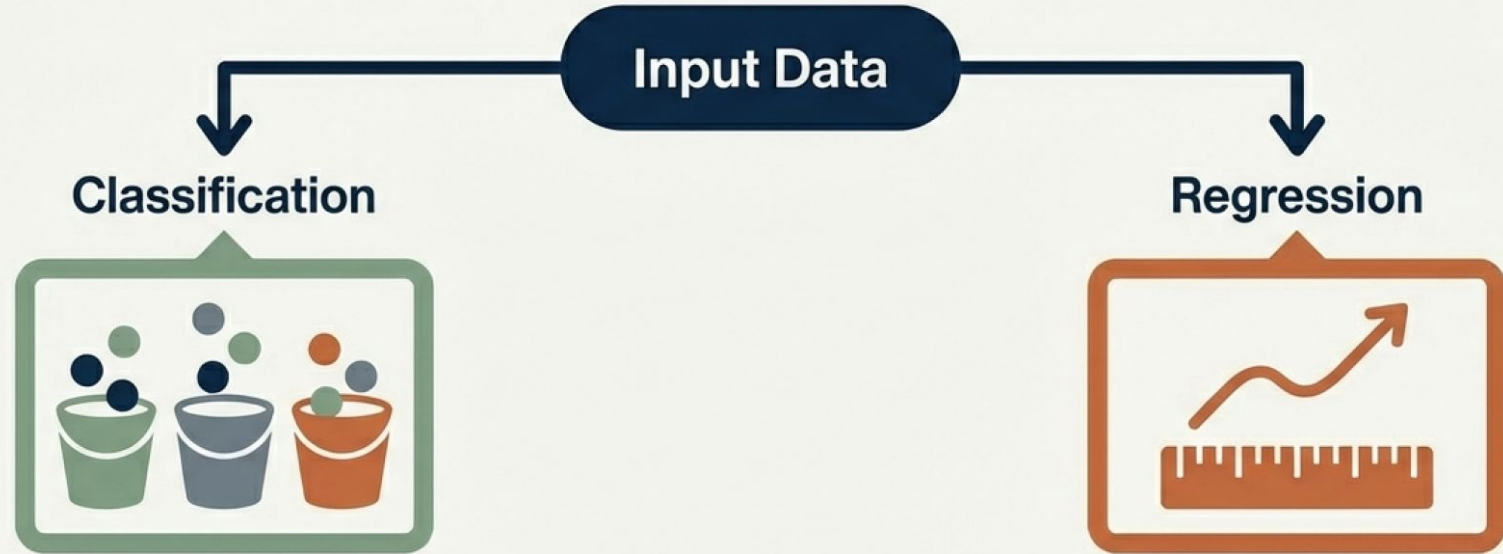


The Algorithm.

- Prediction without evaluation is guessing.

Evaluation Metrics allow us to:
Measure performance • Compare models • Detect overfitting • Improve outcomes.

Types Of Problems



Is the answer a Category?

- Examples: Spam vs. Not Spam, Cancer vs. No Cancer.
- Metrics: Accuracy, Precision, Recall, F1 Score.

Is the answer a Number?

- Examples: House Prices, Salary, Temperature.
- Metrics: MAE, MSE, RMSE, R^2 .

The Foundation: The Confusion Matrix.

Every classification metric stems from these four outcomes.

	Predicted Positive (Sick)	Predicted Negative (Healthy)
Actual Positive (Sick)	True Positive (TP) Sick person correctly identified.	False Negative (FN) Sick person marked healthy. (The Dangerous Miss).
Actual Negative (Healthy)	False Positive (FP) Healthy person marked sick. (The False Alarm).	True Negative (TN) Healthy person correctly identified.

Accuracy

Out of all predictions, how many were correct?

Correct Predictions

True Positive
(TP)

+

True Negative
(TN)

Total Predictions

True
Positive
(TP)

+

True
Negative
(TN)

+

False
Positive
(FP)

+

False
Negative
(FN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}}$$

The Accuracy Paradox.

Why 99% accuracy can be 100% useless.

Scenario: Imbalanced Data (99 Healthy, 1 Sick).
The model predicts 'Healthy' for everyone.

Calculation:
(99 Correct / 100 Total)
= 99% Accuracy.



The Trap:

The model missed the only sick patient. High accuracy masked a complete failure.

Precision: The Cost of False Alarms.



False Positive
(The Boy Who
Cried Wolf).

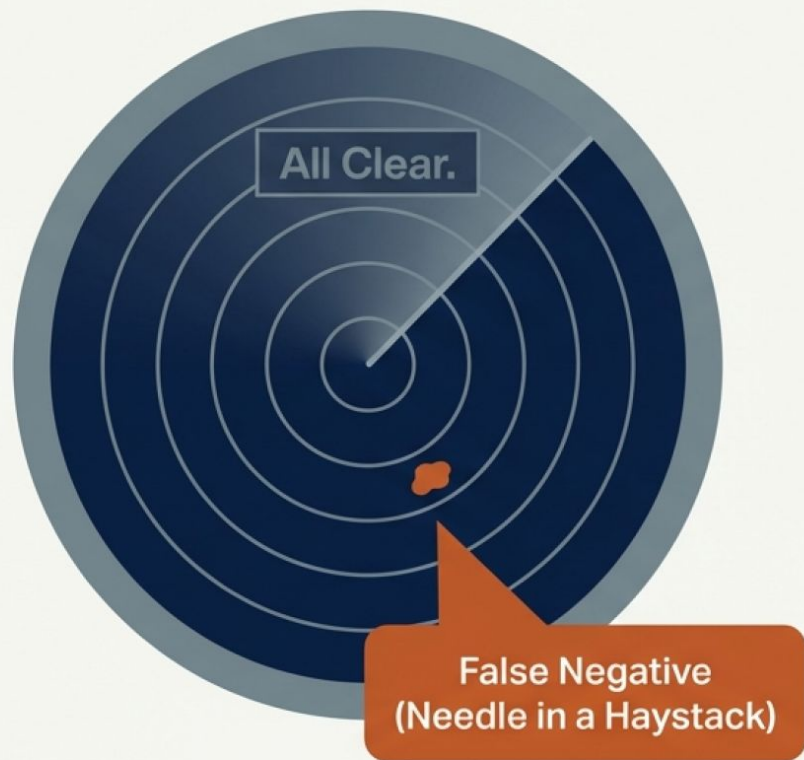
Out of all predicted positives, how many were actually positive?

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

Business Case

- **Scenario:** Spam Filter.
- **Goal:** Catch spam, but never delete a real email.
- **Rule:** When False Positives are annoying or costly, optimize for Precision.

Recall: The Cost of Missing Out.



Out of all actual positives, how many did we catch?

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FN})}$$

Business Case

- Scenario: Cancer Detection.
- Goal: Find every sick patient.
- Rule: When False Negatives are dangerous or fatal, optimize for Recall.

The Great Trade-off: The F1 Score.



Precision and Recall often fight against each other. Improving one usually hurts the other.
The Solution: F1 Score is the harmonic mean, acting as the bridge between the two.

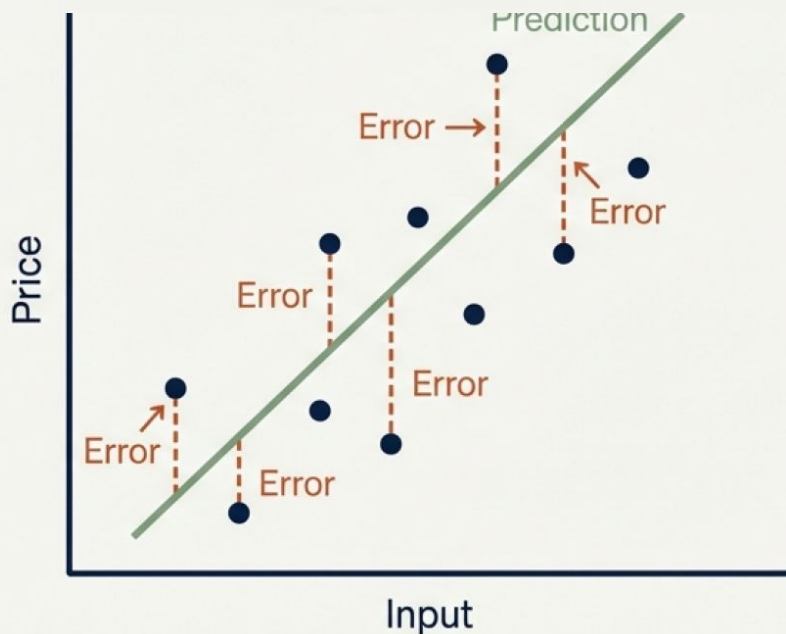
$$F1 = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

When to use

- Imbalanced Datasets (Fraud, Rare Diseases)
- When both False Positives and False Negatives matter equally.

Measuring Distance: Mean Absolute Error (MAE)

Regression Metrics: When the answer is a number



The Logic: How far off are we, on average?

Formula:
Average of
 $|\text{Actual} - \text{Predicted}|$

Example Calculation

Actual Price: \$100 |
Predicted: \$90 | Error: 10

Actual Price: \$150 |
Predicted: \$160 | Error: 10

MAE calculates the average of these absolute errors.

Benefit: Robust and human-readable.

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

- n = number of data points
- y_i = actual value
- $\hat{y}_i$ = predicted value

Example: Given actual values [1, 2, 3] and
predicted values [1.1, 1.9, 3.2],
 $\text{MSE} = (1/3) [(1-1.1)^2 + (2-1.9)^2 + (3-3.2)^2] = 0.02$

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

- n = number of data points
- y_i = actual value
- $\hat{y}_i$ = predicted value

Example: Using the MSE from the previous slide, $\text{RMSE} = \sqrt{0.02} \approx 0.141$

MAE vs. MSE vs. RMSE

MAE (Mean Absolute Error)

 Penalizes Big Errors: No

 Unit: Same

 Robust to Outliers: Yes

MSE (Mean Squared Error)

 Penalizes Big Errors: Very high

 Unit: Squared

 Robust to Outliers: No

RMSE (Root Mean Squared Error)

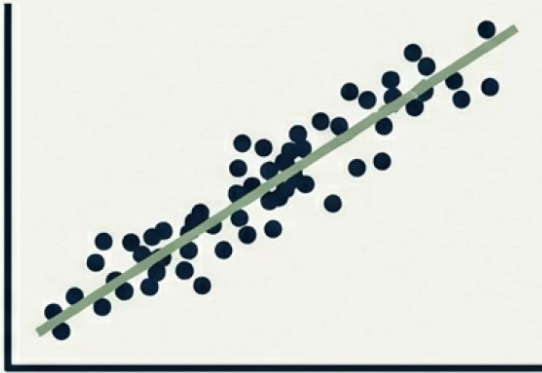
 Penalizes Big Errors: High

 Unit: Same

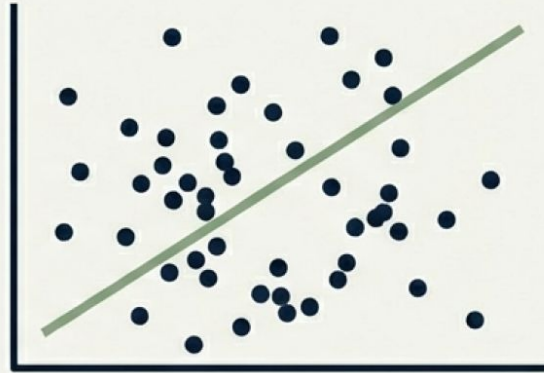
 Robust to Outliers: No

R² Score

High R² (0.95)



Low R² (0.15)



- Concept: How much of the variance in the world does our model explain?
- Scale: 0 to 1 (where 1 is a perfect fit).
- Analogy: Think of R² as a 'grade' for the model's explanatory power, regardless of the unit of measurement.

R² Formula & Score Range

R² Formula

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

SS_{res} (Sum of Squared Residuals): The sum of the squared differences between the actual values and the predicted values.

SS_{tot} (Total Sum of Squares): The sum of the squared differences between the actual values and the mean of the actual values.

R² Score Range

< 0 (Negative)



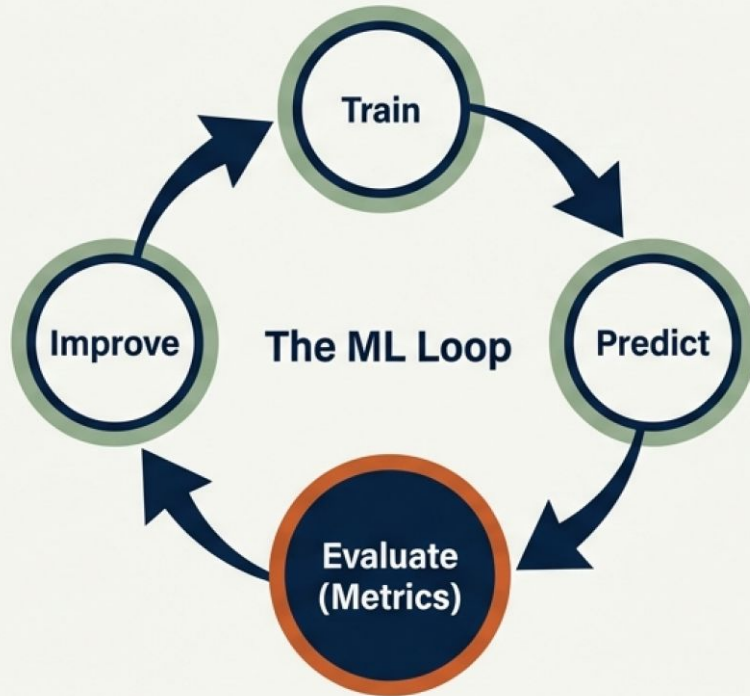
Worse than the mean model.
Model is arbitrarily poor.

No better than the mean model.
Explains none of the variance.

Perfect fit. Explains 100%
of the variance.

Typical range is 0 to 1. Higher is generally better, indicating more variance is explained by the model.

From Evaluation to Optimization.



Metrics are the compass for the "Improve" stage.

Common Pitfalls to avoid:

- Relying solely on Accuracy.
- Ignoring Class Imbalance.
- Forgetting the business context.

**"A model is not judged by its code, but by its impact.
Metrics are the language of impact."**

